

longer mattered. Uncertainty seemed to be banished. The Nasdaq doubled over six months, followed by an epic crash.

After that implosion, we saw the opposite extreme: Profitable, debt-free tech companies were being traded for less than book value, and in a few rare instances, they were being sold for *less than cash on hand*. Investors had become certain that a dollar was worth only 75 cents.

There was little uncertainty heading into the March 2009 stock-market lows. Almost everyone was sure the world was falling into the abyss. In that massive and indiscriminate selling, it seemed almost certain that no one was ever going to buy another house or car, send their kids to school, or for that matter, clothe and feed them. How did the consensus work out in that instance? (Not well.)

When we discuss uncertainty, what we are really discussing is the unknown. All unknown outcomes contain risk, and therein lies the possibility of loss. Risk is inherent in the concept of uncertainty. However, anyone looking for performance must embrace risk, for without it, there can be no reward.

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Michael Mauboussin, the original thinker we met in earlier chapters, points out that people sometimes confuse *risk* with *uncertainty*.

Risk, he observes, occurs when we don't know what is going to happen next, but we do know *what the distribution set looks like*. Uncertainty, on the other hand, occurs when we have no idea what is going to happen next and *we do not know what the possible distribution looks like*.

In other words, the future is always unknown, but that does not make it uncertain. Rather, we should quantify this in the language of *statistical probability*.

When we don't know what a future outcome will be but understand the probability distribution—think of dice or a multiple-choice exam—we have risk, but we do not have uncertainty. We never know what the roll of the die will be in advance, but we do know it will be one of six possibilities.